

RAVEN ROCK, PA, USA

WMO: 722826

Lat: 39.733N

Lon: 77.433W

Elev: 902

StdP: 14.22

Time Zone: -5.00 (NAE)

Period: 06-17

WBAN: 93762

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.8	15.1	-4.9	4.4	13.8	-1.1	5.3	17.6	17.3	34.3	15.9	33.0	4.7	230	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	92.1	73.2	89.1	71.8	86.5	70.7	76.3	87.2	74.9	84.8	73.5	82.6	4.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.0	126.7	81.4	71.7	121.3	80.3	70.4	115.7	79.3	40.3	87.3	38.9	84.9	37.6	82.9	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.8	11.8	10.3	DB	6.6	97.4	5.2	6.7	2.9	102.2	-0.2	106.1	-3.1	109.9	-6.9	114.8
			WB	4.9	78.7	4.3	2.1	1.8	80.3	-0.7	81.5	-3.2	82.7	-6.3	84.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.7	31.8	33.6	42.9	54.0	63.3	71.5	75.9	74.3	67.2	56.8	46.2	37.9
	DBStd	17.44	9.96	11.06	10.56	9.73	7.96	5.90	5.68	4.86	6.86	8.15	8.31	9.19
	HDD50	1967	573	470	271	64	2	0	0	0	0	28	167	392
	HDD65	4833	1029	878	687	354	133	14	1	2	55	277	564	839
	CDD50	3692	9	11	50	183	414	644	802	753	516	239	53	18
	CDD65	1085	0	0	2	23	80	207	338	291	121	23	0	0
	CDH74	9262	0	3	13	227	668	1705	3186	2400	917	142	0	1
	CDH80	3115	0	0	0	46	171	512	1276	796	286	28	0	0
Wind	WSAvg	3.0	4.0	4.2	4.2	4.0	2.5	2.3	2.0	1.6	1.8	2.7	3.2	3.5
Precipitation	PrecAvg	44.0	3.3	2.5	4.1	3.7	3.7	4.0	3.9	3.8	4.6	3.4	3.1	3.5
	PrecMax	68.9	8.3	5.9	8.9	7.6	8.4	8.7	9.6	6.0	9.6	7.6	6.7	6.3
	PrecMin	28.6	1.2	0.3	0.5	1.7	0.9	1.3	1.7	1.5	0.0	0.5	0.5	0.6
	PrecStd	9.1	1.8	1.3	1.8	1.5	1.9	2.2	1.9	1.2	2.7	2.0	1.5	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.9	67.9	75.6	85.5	88.7	92.4	99.0	93.3	91.8	83.8	71.0	67.3
		MCWB	56.3	52.9	58.5	62.7	69.4	74.3	76.5	72.8	72.4	68.2	58.4	59.0
	2%	DB	59.4	60.7	69.7	80.2	84.7	88.4	93.4	89.9	87.2	77.7	66.4	61.8
		MCWB	53.0	48.0	55.6	61.2	68.5	71.8	73.7	73.0	70.8	64.9	57.0	56.2
	5%	DB	52.4	55.5	64.6	75.7	81.1	85.6	90.0	87.1	83.0	73.6	62.8	56.8
		MCWB	46.8	44.5	52.8	59.2	66.3	70.7	72.0	72.0	69.3	63.9	53.9	51.6
	10%	DB	46.3	49.6	59.8	70.9	77.5	82.7	86.7	84.7	79.3	69.9	59.3	51.1
		MCWB	40.0	41.2	50.4	57.4	63.9	68.4	70.9	70.4	67.0	60.9	51.2	44.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.8	56.3	60.8	67.4	73.6	76.5	78.8	78.2	75.3	71.5	62.3	61.7
		MCDB	62.5	65.2	70.4	77.0	85.2	88.0	92.7	88.8	85.5	77.3	68.1	65.2
	2%	WB	53.4	49.4	58.4	64.2	71.1	74.5	76.8	76.0	73.4	68.2	58.6	56.7
		MCDB	57.1	58.5	66.3	76.1	81.7	85.5	88.8	85.2	82.9	76.3	64.1	61.1
	5%	WB	46.4	45.7	55.3	61.4	69.1	72.8	75.1	74.4	71.5	65.4	55.5	52.4
		MCDB	52.8	55.3	63.3	72.2	78.1	82.7	85.7	82.5	80.1	71.3	61.8	56.1
	10%	WB	40.1	41.4	50.6	58.8	67.0	71.2	73.7	72.7	69.4	62.4	52.8	45.9
		MCDB	46.4	48.6	59.2	69.0	74.5	79.9	83.2	80.8	76.3	67.9	57.9	50.0
Mean Daily Temperature Range	5% DB	MDBR	13.2	16.0	17.6	18.8	18.7	18.5	19.0	18.6	18.7	17.3	15.5	13.2
		MCDBR	20.4	24.8	24.3	25.1	22.7	20.6	22.0	20.8	22.4	22.0	20.1	19.5
		MCWBR	16.3	17.0	15.1	12.7	11.2	10.1	8.4	8.5	9.9	11.3	12.9	15.5
	5% WB	MCDBR	19.9	22.5	21.7	22.0	20.2	18.7	18.6	17.6	19.2	18.4	17.1	19.2
		MCWBR	17.2	16.7	15.3	12.8	10.5	10.1	8.3	8.5	9.4	11.3	13.0	16.7
Clear-Sky Solar Irradiance	taub	0.304	0.319	0.351	0.393	0.429	0.494	0.512	0.505	0.425	0.361	0.318	0.304	
	taud	2.437	2.376	2.369	2.273	2.202	2.056	2.023	2.022	2.223	2.408	2.493	2.473	
	Ebn at Noon	274	285	286	280	271	252	246	244	258	266	267	266	
	Edh at Noon	26	31	35	41	45	53	54	53	40	30	24	23	
All-Sky Solar Radiation	RadAvg	595	878	1168	1506	1686	1869	1855	1646	1342	975	692	516	
	RadStd	50	104	75	127	184	148	108	86	175	110	62	56	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (64 neighbors)		+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+216	+121

Nomenclature: See separate page